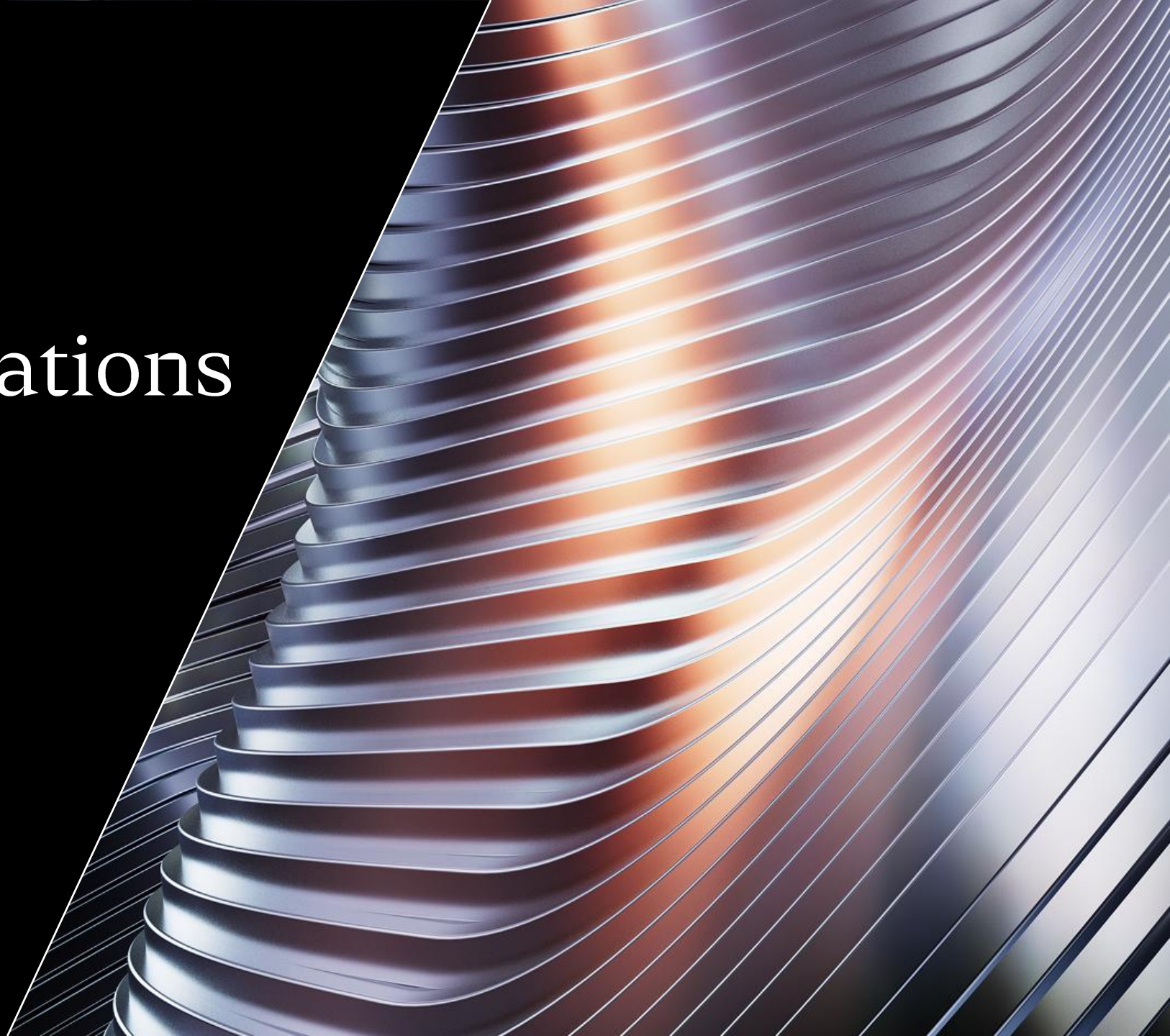




Unsupervised Learning Foundations and Clustering

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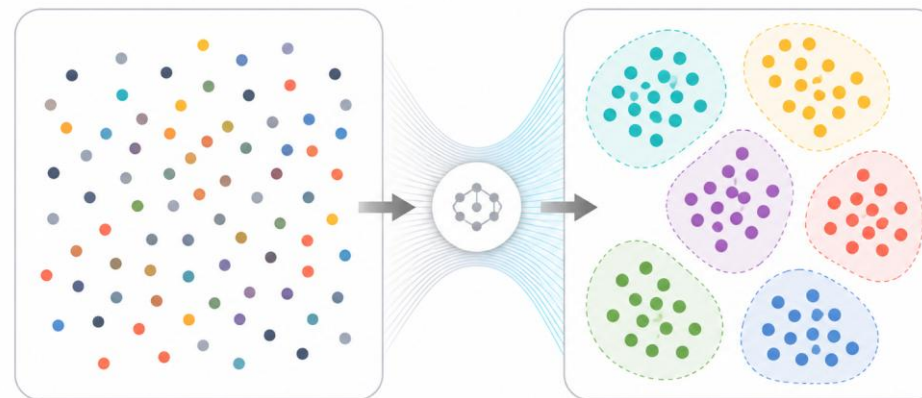
LEARNING OBJECTIVES

- Explain what unsupervised learning is.
 - Distinguish supervised, unsupervised, semi-supervised, and reinforcement learning.
 - Describe clustering and association as major unsupervised learning tasks.
 - Explain hard vs. soft clustering.
 - Compare K-Means, K-Medoids, DBSCAN, Hierarchical Clustering, and GMM.
 - Identify appropriate clustering use cases and limitations.
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UNSUPERVISED LEARNING: FINDING STRUCTURE WITHOUT LABELS

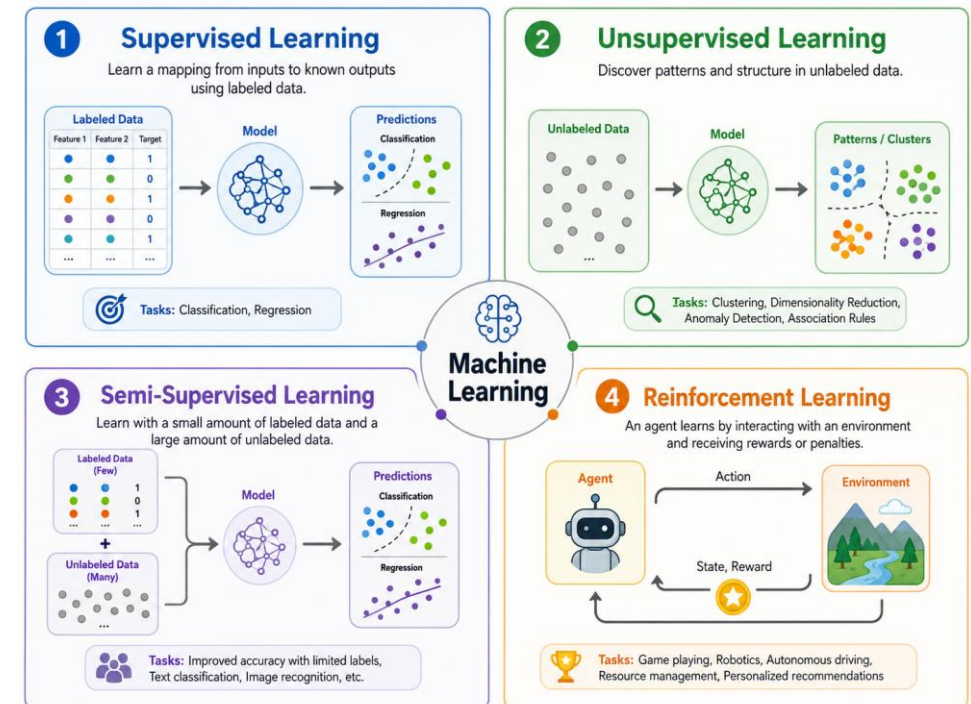
- Discovering patterns when no target variable exists
- From raw data to groups, anomalies, and hidden relationships





WHERE UNSUPERVISED LEARNING FITS

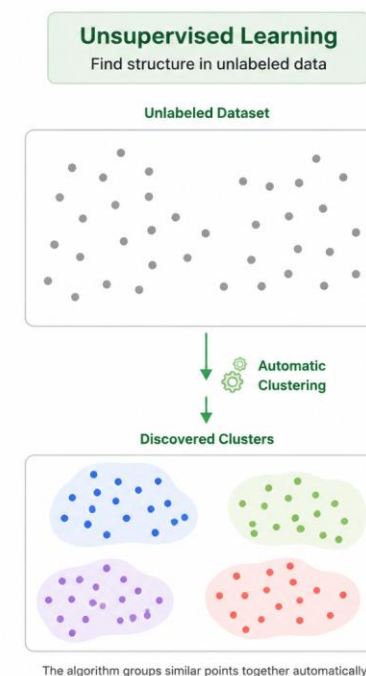
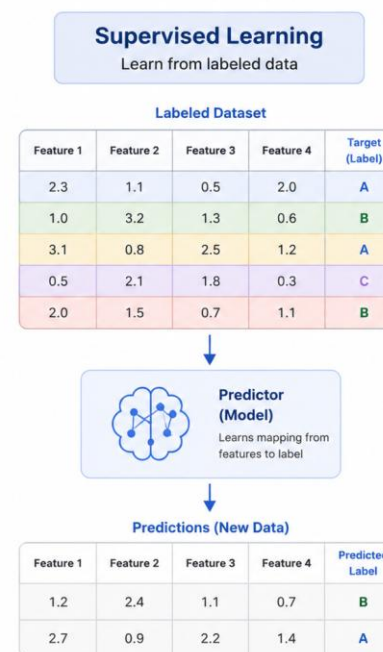
- Machine Learning can be grouped into:
 - Supervised learning
 - Unsupervised learning
 - Semi-supervised learning
 - Reinforcement learning
- Unsupervised learning uses unlabeled data
- The goal is not prediction, but pattern discovery





SUPERVISED VS. UNSUPERVISED LEARNING

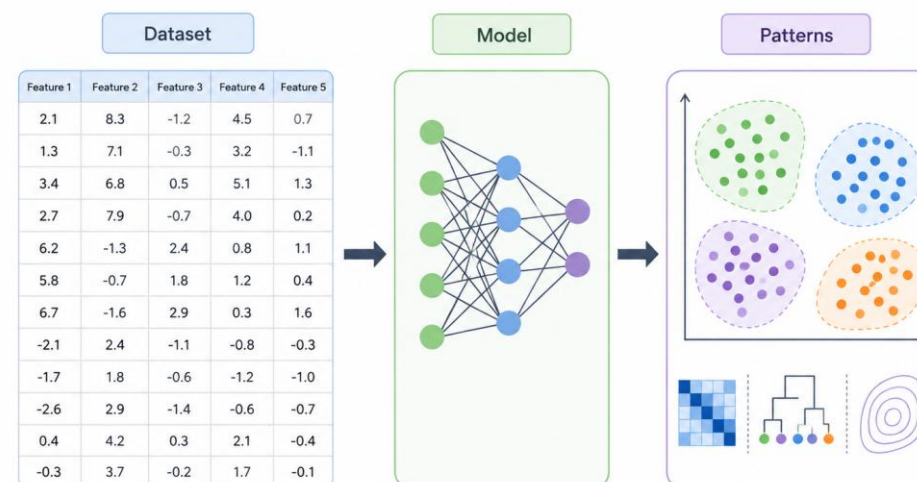
Supervised Learning	Unsupervised Learning
Has input and output labels	Has input data only
Learns to predict	Learns to discover structure
Classification, regression	Clustering, association, dimensionality reduction
Example: spam vs. not spam	Example: group customers by behavior





WHAT IS UNSUPERVISED LEARNING?

- A model analyzes data without predefined labels
- It searches for:
 - Similarities
 - Differences
 - Hidden patterns
 - Natural groupings
- Useful when the dataset has no clear target variable





WHY USING UNSUPERVISED LEARNING?

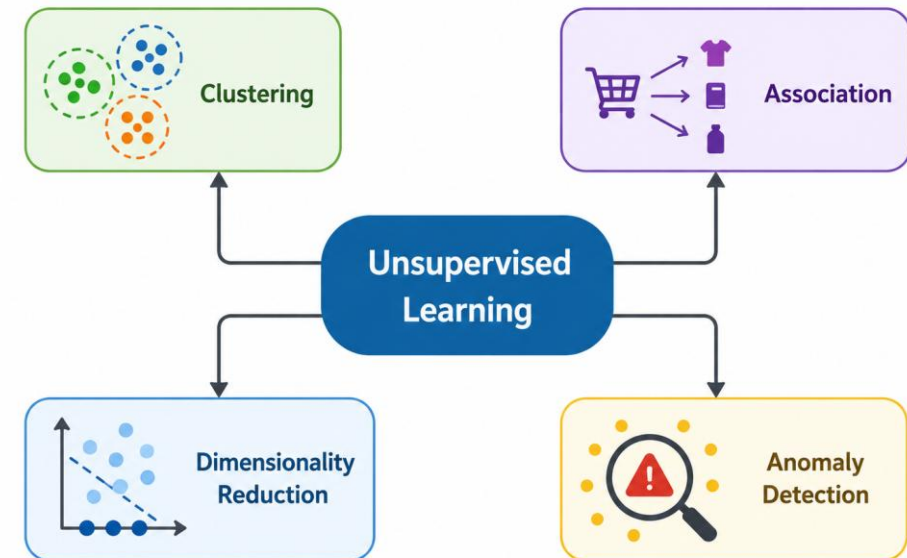
- Real-world data is often unlabeled
- Labeling data can be expensive and slow
- Helps with:
 - Exploration
 - Data understanding
 - Segmentation
 - Anomaly detection
 - Preprocessing before supervised learning





MAIN TYPES OF UNSUPERVISED LEARNING

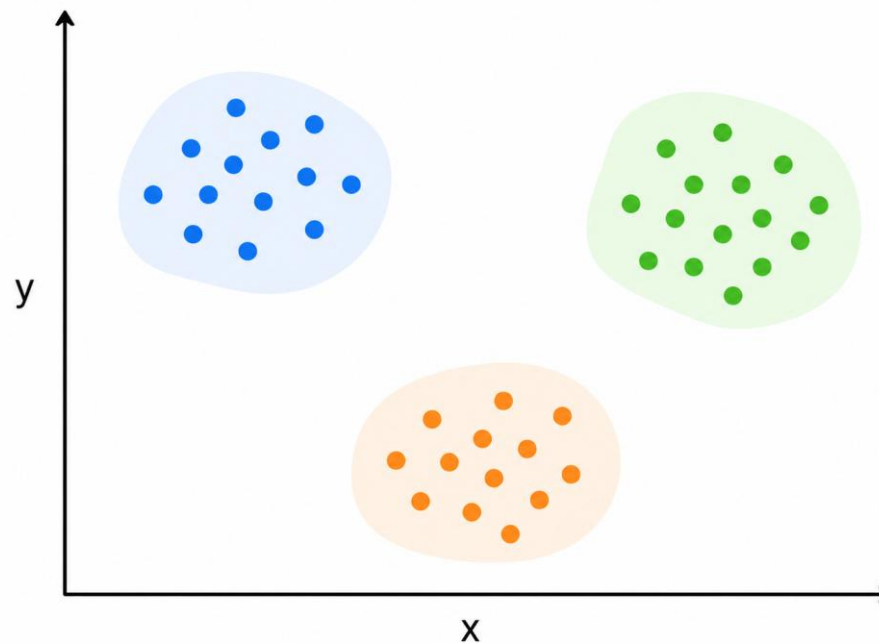
- **Clustering:** Groups similar data points
- **Association:** Finds relationships between items or events
- **Dimensionality reduction:** Compresses features while preserving useful information
- **Anomaly detection:** Finds unusual behavior or outliers





WHAT IS CLUSTERING?

- Clustering groups similar data points together
- Points inside a cluster should be more similar to each other
- Points in different clusters should be more different
- Similarity can be based on:
 - Euclidean distance
 - Manhattan distance
 - Cosine similarity





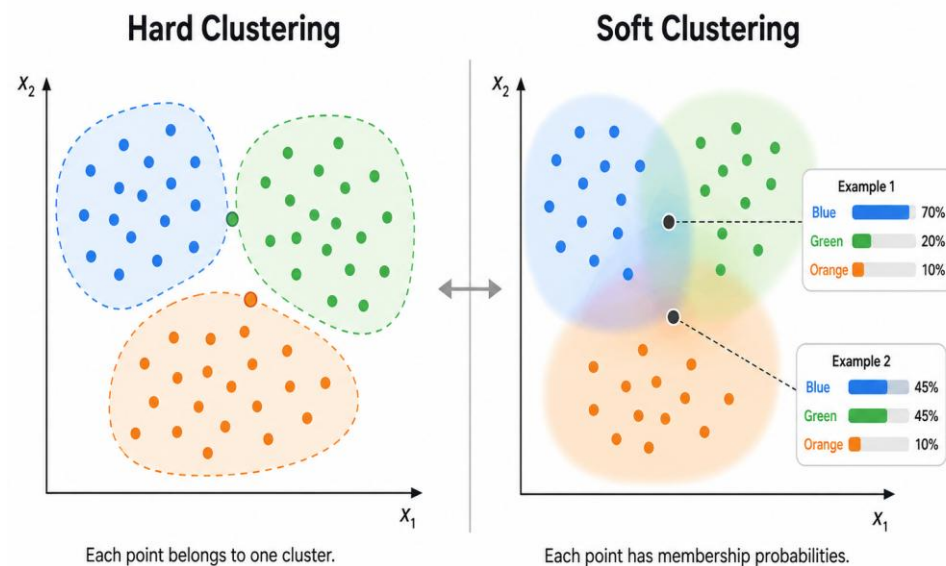
HARD VS. SOFT CLUSTERING

Hard clustering

- Each point belongs to exactly one cluster
- **Example:** customer A belongs to segment 1

Soft clustering

- Each point has probabilities of belonging to multiple clusters
- **Example:** customer A is 70% segment 1, 30% segment 2



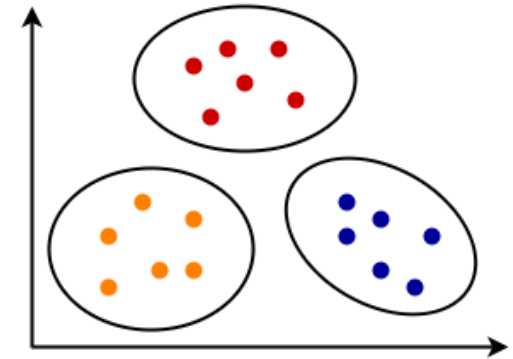


K-MEANS CLUSTERING

- Centroid-based clustering
- Requires choosing the number of clusters: k
- Repeats two steps:
 1. Assign points to nearest centroid
 2. Move centroids to the center of assigned points
- Works best with compact, spherical clusters



Before K-Means

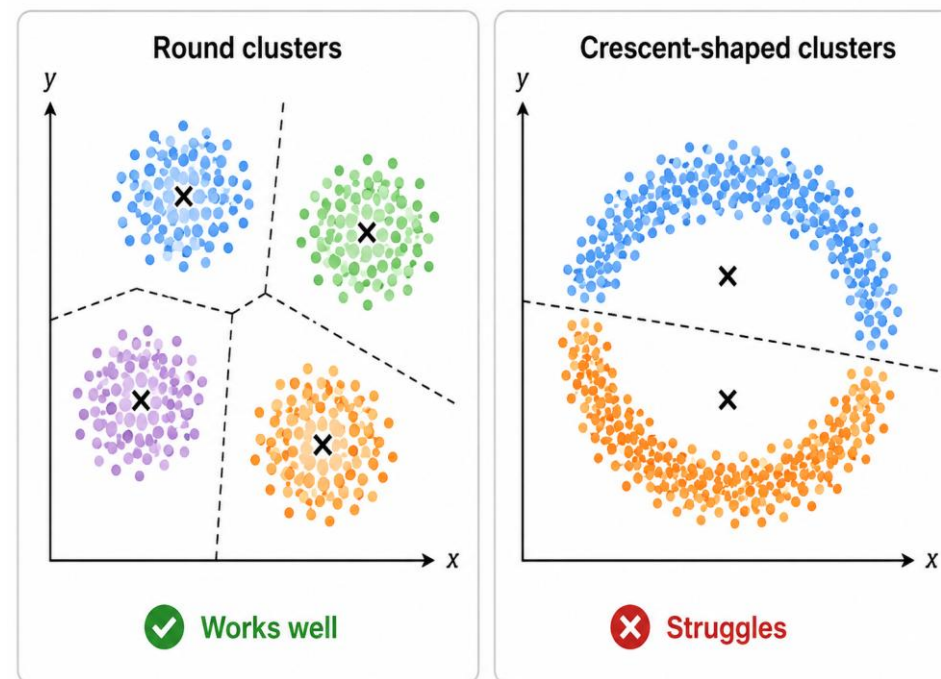


After K-Means



K-MEANS: STRENGTHS AND WEAKNESSES

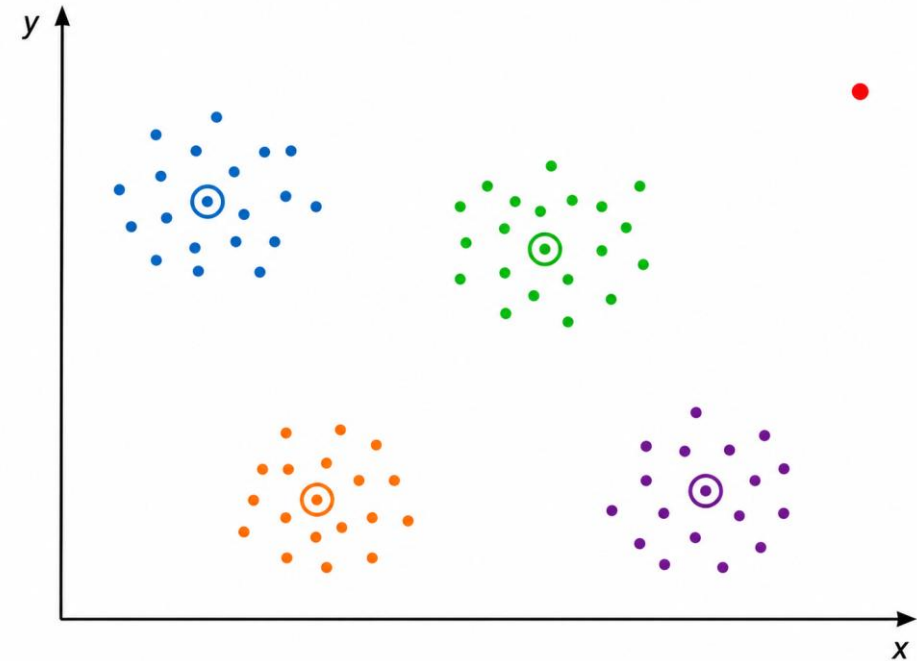
Strengths	Weaknesses
Simple and intuitive	Requires predefined k
Fast and scalable	Sensitive to initialization
Easy to visualize	Sensitive to outliers
	Struggles with irregular cluster shapes





K-MEDOIDS

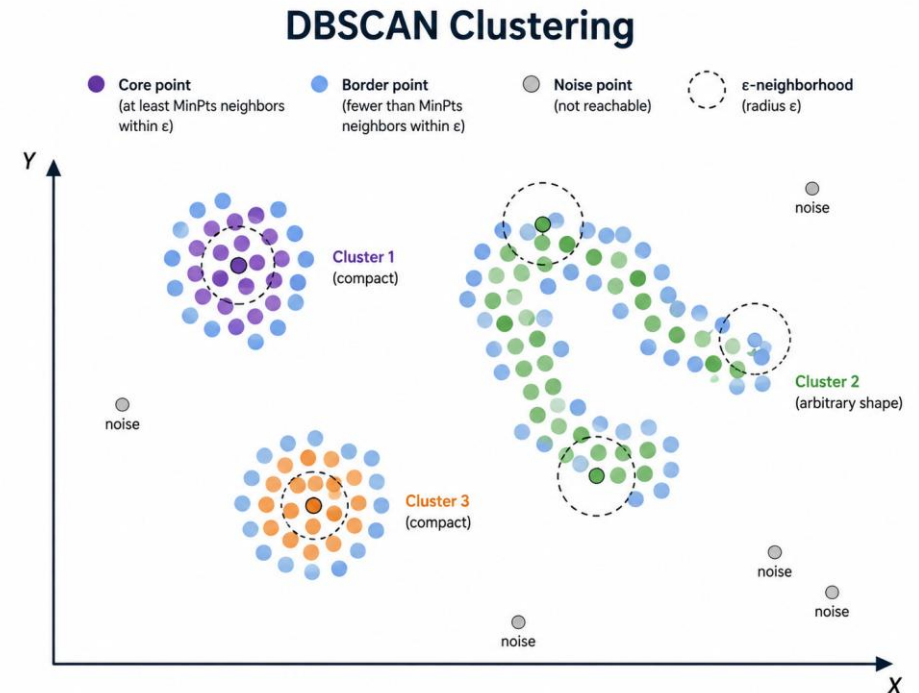
- Similar to K-Means
- Uses real data points as centers: medoids
- More robust to outliers
- Can support different distance metrics
- Usually slower than K-Means





DBSCAN

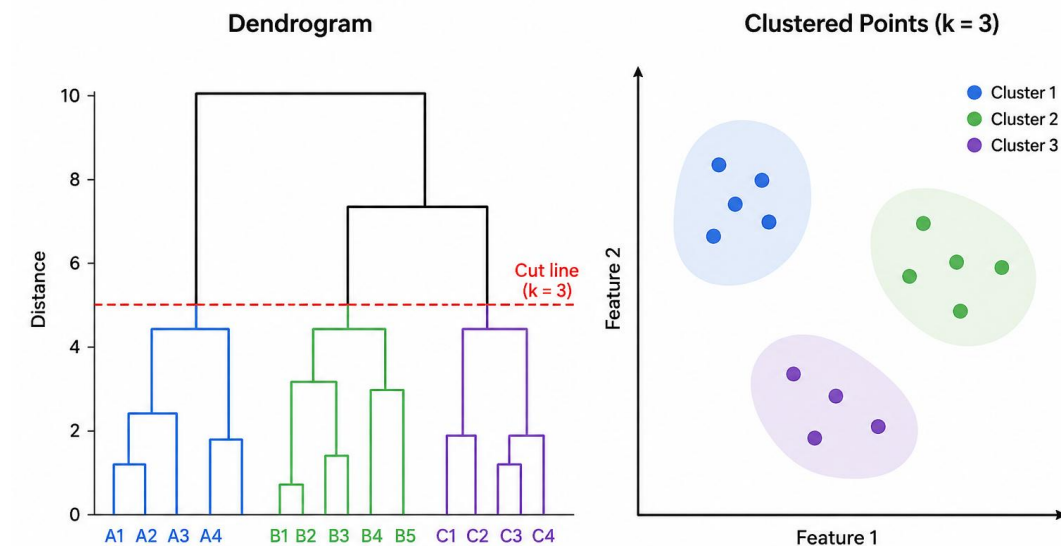
- Density-based clustering
- Finds dense regions of points
- Detects noise and outliers
- Does not require predefined k
- Main parameters:
 - *epsilon*: neighborhood radius
 - *MinPts*: minimum points for dense region





HIERARCHICAL CLUSTERING

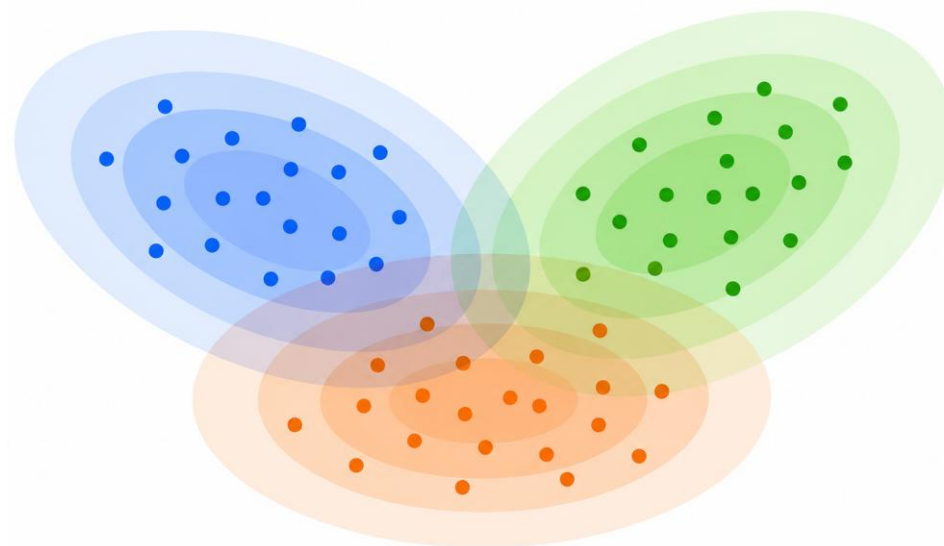
- Builds a hierarchy of clusters
- Two main approaches:
 - **Agglomerative**: bottom-up merging
 - **Divisive**: top-down splitting
- Often visualized with a dendrogram
- Useful for understanding nested relationships





GAUSSIAN MIXTURE MODELS

- Distribution-based clustering
- Assumes data comes from multiple probability distributions
- Produces soft cluster assignments
- Useful when clusters overlap
- More flexible than K-Means for elliptical clusters





CHOOSING A CLUSTERING METHOD

Situation	Good option
Large, clean, spherical clusters	K-Means
Outliers are important	K-Medoids or DBSCAN
Irregular shapes	DBSCAN
Need hierarchy	Hierarchical clustering
Overlapping groups	GMM



APPLICATIONS OF CLUSTERING

- Customer segmentation
 - Fraud detection
 - Medical image analysis
 - Social network communities
 - Cybersecurity behavior analysis
 - Traffic and city planning
 - Document grouping
 - Product recommendation support
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KEY TAKEAWAYS

- Unsupervised learning discovers structure without labels
 - Clustering is one of its most important tasks
 - Different clustering algorithms make different assumptions
 - No clustering result is automatically “correct”
 - Human interpretation is essential
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Technology Elevated